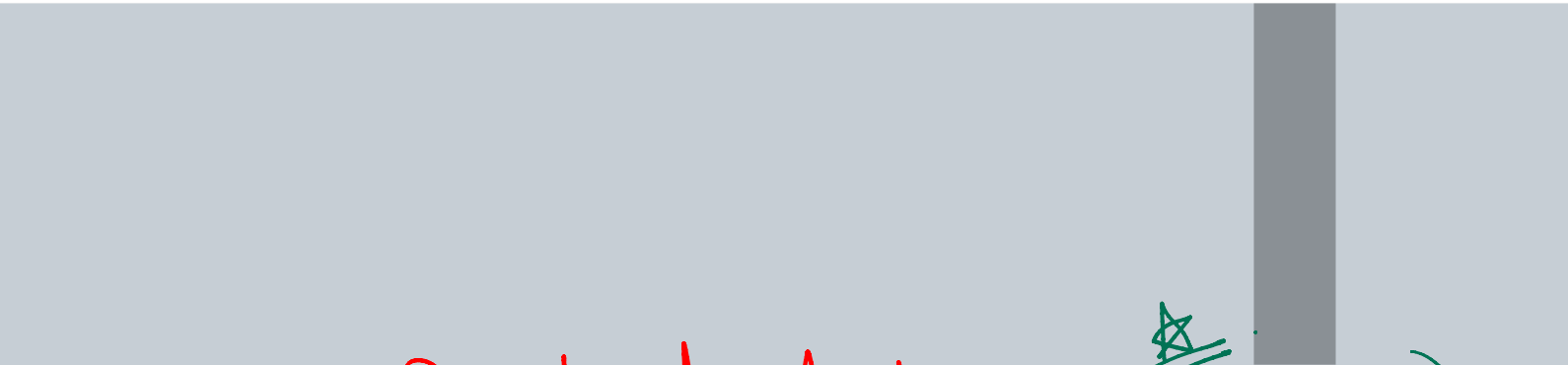
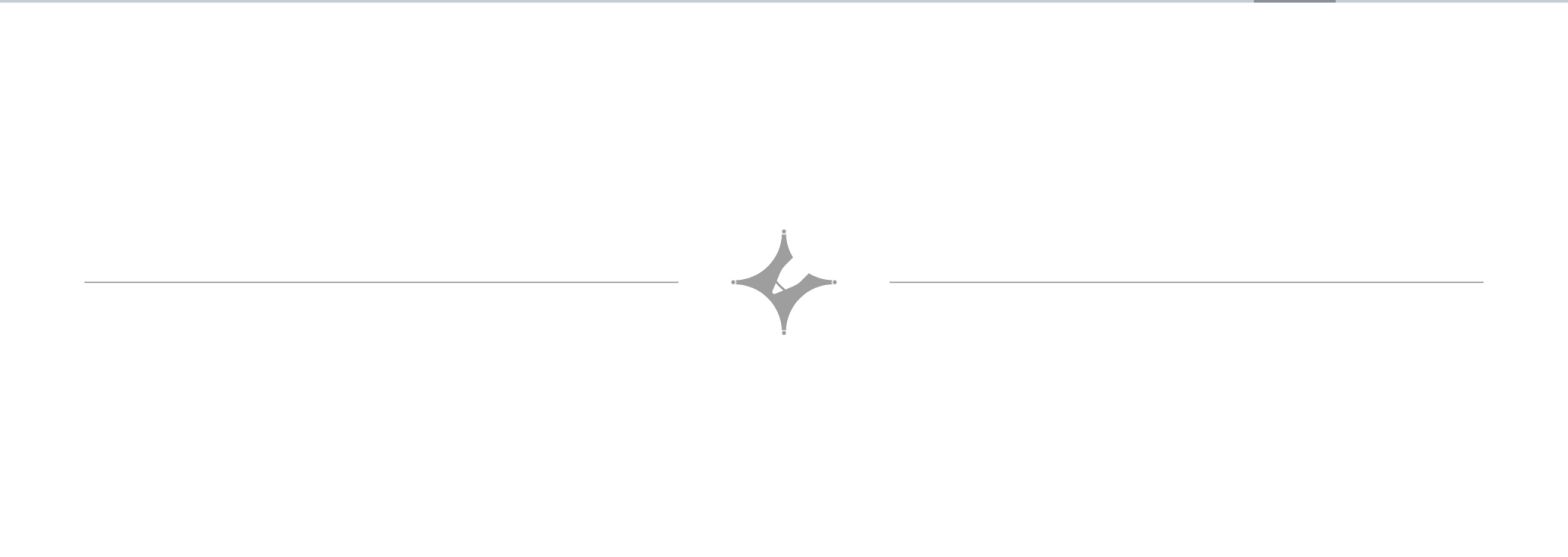
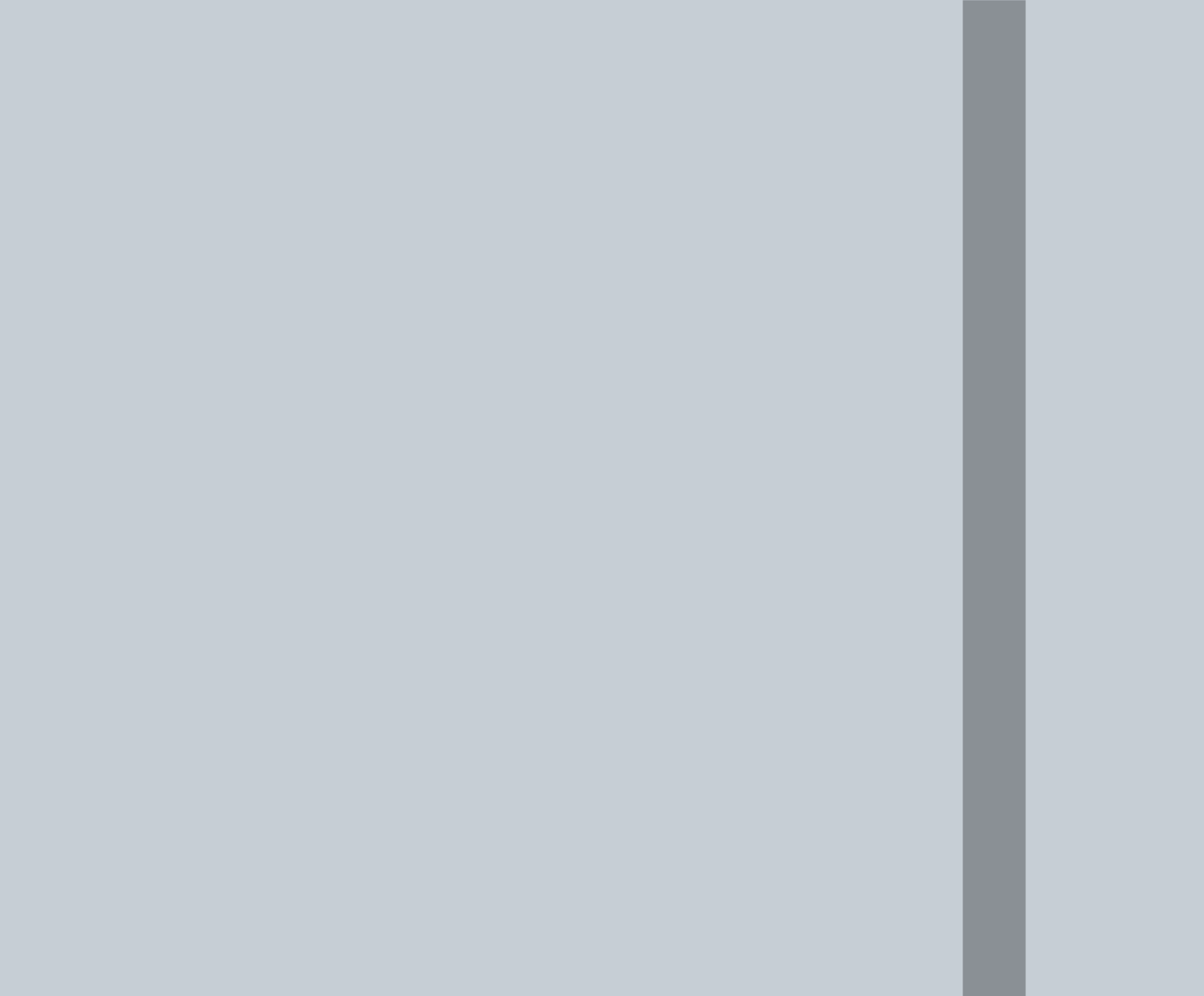
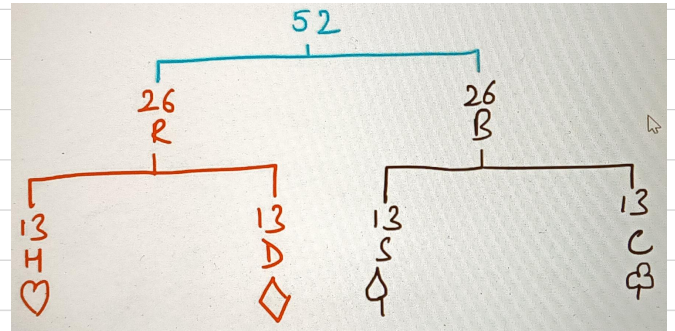




(P) Probability

or $\rightarrow +$
and $\rightarrow \times$

$$P = \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes}}$$



3. Two cards are drawn from a well-shuffled pack of cards. What is the probability that the first is a spade and the second is a diamond?

Ans: _____

$$P = P(S, D) = \frac{13}{52} \times \frac{13}{51} = \frac{13}{204}$$

4. Two cards are drawn from a well-shuffled pack of cards. What is the probability of getting a spade and a diamond?

Ans: _____

$$P = P(S, D) \text{ or } P(D, S)$$

$$= \frac{13}{52} \times \frac{13}{51} + \frac{13}{52} \times \frac{13}{51} = \frac{2 \times 13}{2 \times 51} = \frac{13}{102}$$

5. Two bottles are randomly selected from a stack of 10 bottles in which 5 are blue, 3 are green, and 2 are yellow. What is the probability that the 1st bottle selected is blue and the 2nd is green?

Ans: _____

$$P = P(B, G) = \frac{5}{10} \times \frac{3}{9} = \frac{1}{6}$$

6. Three bottles are randomly selected from a stack of 12 bottles in which 3 are black, 4 are white, and 5 are red. What is the probability that the selected are of different colour?

Ans: _____

$$P = P(B, W, R) = \frac{3}{12} \times \frac{4}{11} \times \frac{5}{10} = \frac{1}{22}$$

$$\frac{1}{22} \times 3! = \frac{3 \times 2}{22} = \frac{3}{11}$$

agar mention hota ki 1st ball is black, 2nd white etc to 1/22 ans hota but mention nhi that to there are 3! ways to arrange 3 thing (P & C me para).

7. Two dice are rolled. What is the probability that the sum of the results is 5?

Ans: _____

(Khud se solve kar liya tha mere but fir bhi notes me dad diya just for precaution)

$$T = 6 \times 6 = 36$$

F	D ₁	D ₂
	1	4
	2	3
	3	2
	4	1
	5	X
	6	X

$$P = \frac{4}{36} = \frac{1}{9}$$

9. A fair coin is tossed 6 times. What is the probability that it turns up exactly 2 times?

Ans: _____

$$T = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6 = 64$$

$$F = HH TTTT \quad \frac{6!}{2! \times 4!} = \frac{6 \times 5}{2} = 15$$

$$P = \frac{15}{64}$$

10. A bag contains three differently coloured bottles, which include 3 black, 2 white, and 5 red. If 3 bottles are picked randomly from the bag, what is the probability that:

i. All the three are black? Ans: _____

ii. None of them are white? Ans: _____

iii. All of them are not white? Ans: _____

$$i) P(B, B, B) = \frac{3}{12} \times \frac{2}{11} \times \frac{1}{10} = \frac{1}{220}$$

$$ii) P = \frac{8}{12} \times \frac{7}{11} \times \frac{6}{10} = \frac{14}{55}$$

$$iii) P(All w) = \frac{4}{12} \times \frac{3}{11} \times \frac{2}{10} = \frac{1}{55}$$

$$P(All Not w) = 1 - \frac{1}{55} = \frac{54}{55}$$

→ koi bhi ball white nahi ho sakti.

→ 1 ya 2 ya 0 white ho sakti hai
teeno white nahi honi chahiye

that's the difference.

11. A committee of 10 people needs to be seated on 10 chairs in a straight line. What is the probability that 3 particular people always sit together?

Ans: _____

$$T = 10!$$

$$F = 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ \boxed{8 \ 9 \ 10}$$

$$8! \times 3!$$

$$P = \frac{8! \times 3!}{10!} = \frac{3 \times 2}{10 \times 9} = \frac{1}{15}$$

